

# Shipaton 2026 submission answers

Next Gen award. Fields are in the same order as the Devpost form. Copy the text inside each shaded box. Anything in [square brackets] is for you to fill in. Fields marked LEAVE BLANK stay empty; the note isn't for pasting. The same answers are in Root-Shipaton-Answers.txt.

PROJECT DETAILS (public project page)

## 1. About the project

**PASTE** · Paste as-is; it is already Markdown. In the PDF it spans 7 boxes (copy them in order); in the TXT it is one block.

### ## Inspiration

We have always been drawn to cultures and languages. And we kept meeting the same person in different forms: someone who can name every dish at a family gathering, who laughs at the right moment in a story, who recognises their grandmother's greeting the second they hear it, but who can't answer back in their own indigenous language.

Often it isn't because they didn't care. It's because they grew up away from the place where that language lives every day: in a city, in another country, in a school that taught in English. The language didn't disappear. It just got quieter with every mile.

The big language apps rarely teach these languages, and when they do, they are built for tourists: order a coffee, find the hotel. A heritage learner wants something else. They want to greet an elder properly. To understand the proverb at a wedding. To say \*karibu\* and mean it.

So we built **Root**, around one belief:

> **Distance from home shouldn't become distance from your language.**

### ## What it does

Root is a calm Android app for reconnecting with a heritage language, starting with **Dholuo, Shona, Swahili and Amharic**. It is a session, not a destination: open it, bring one word back to mind, and leave a little closer to home.

- **Practice** opens straight onto a recall card. "Take a breath. The word is somewhere in you." Reveal it, rate yourself honestly (Missed, Close or Got it), and watch a root grow, but only for the words you truly remembered. No streaks, no leaderboards, no guilt.
- **Learn** gives every language a gentle path of guided conversations, pattern workshops and stories. Nothing is locked, and a lesson you finish stays finished.
- **Explore** is language as it's lived: everyday **Situations** (greetings, meeting people, the market, getting around), **Stories & culture** (welcoming new neighbours in Mombasa, finding the way in Addis Ababa, why the Ethiopian clock starts at dawn), and **My notebook** for the words worth keeping.
- **Your family's words**. Add a phrase your family actually uses, record a reference with their permission, then record yourself and compare. Your recordings stay on your phone.
- **Real voices only**. Where a genuine recording exists you hear it, with the speaker credited by name. Where one doesn't, Root says so. We never fake a voice with text-to-speech.
- Works offline with no account, in warm-paper light and warm-charcoal dark themes, with a home-screen widget and a shareable card to teach someone one word.

## ## How we built it

Root is fully native: **Kotlin** and **Jetpack Compose** with Material 3. A versioned **Room** database keeps everything on the device, including recall history, lesson attempts, saved phrases and recordings, with tested migrations so updates keep a learner's progress. WorkManager downloads content packs that are size-bounded, hash-verified and only activated once complete. Glance powers the widget.

Content runs through a **Python toolchain** that validates versioned lesson packs and tracks where every line came from. Each starter phrase (39 to 40 per language) was checked against a public reference such as Wikivoyage, Omniglot, Wiktionary or the public-domain FSI Amharic course, and the sources ship inside the app.

**RevenueCat** powers Root Premium, and we designed it around who our learners are. Most heritage learners want *one* language: the one their family speaks. So Premium is a one-time unlock for a single language (US\$4.99) or for every language (US\$9.99), and never a subscription. It opens two extra phrase sets per language, like **Heart words** (\*I love you\*, \*long time no see\*, \*get well soon\*) and **Keep talking** (\*please say that again\*, \*speak slowly\*), the words you reach for with family. Each language has its own entitlement ('premium\_<language>'), and 'premium' is the all-languages bundle. Purchases run through RevenueCat's Test Store for this showcase, entitlements are cached for offline use, restore is built in, and a language is only sold once it has premium sets in the app.

The design is deliberately quiet. Source Serif 4 carries the words and Inter the interface. One hand-drawn root path becomes the logo, the launch animation and your progress, and an original startup sound is scored to that animation.

## ## Challenges we ran into

- **Honesty over volume.** It would have been easy to fill the app with machine translations and synthetic voices. We refused. Finding recordings we could legally use, that matched the exact word, was slow: today six phrases play genuine Lingua Libre recordings, each credited to its volunteer speaker.
- **Earning money without betraying the mission.** We would not put practice behind a paywall. Every starter set stays free. Premium adds more sets on top, and nothing is sold before it exists in the app.
- **Progress you can trust.** At one point, reopening a finished lesson quietly reset it. We rebuilt completion as durable history, so reviewing never erases what you achieved.
- **Four languages, one switch.** Changing language had to change Practice, Learn and Explore together, without losing anything in the language you left behind.

## ## Accomplishments that we're proud of

- A complete, working app in four African languages: over 220 source-checked phrases, 8 Learn units with 24 lessons, 8 short stories and 8 sourced culture notes.
- A monetisation model that respects the people it is for.
- A design that lets the language lead: no confetti, no pressure, just the word.
- Over 350 automated tests across the app, real devices and the content pipeline.

## ## What we learned

- The technology is the easy part of a language app. Trust is the hard part: every source, recording and credit matters.
- Heritage learners don't need to be pushed. They need a doorway, and a reason to come back.
- Monetisation design is product design. Choosing per-language pricing reshaped our entitlements, our content model and our paywall.

## ## What's next for Root

- Work with native speakers, starting with elders, to review every phrase and record it with consent and credit.
- More premium sets, starting with **Talking with elders**: the respectful forms for greeting and visiting family.
- Proverbs and longer spoken stories.
- More languages, led by what learners ask for.
- Publish Root on Google Play.

Because a language survives in the small moments: a greeting at the door, a word at the table, an answer given back in the language it was asked in.

## 2. Built with

**PASTE** · 19 of the 25 allowed tags. Type or paste them one at a time and press Enter after each.

android  
kotlin  
jetpack-compose  
material-design-3  
revenuecat  
room  
sqlite  
kotlinx-serialization  
workmanager  
jetpack-glance  
okhttp  
python  
gradle  
junit  
android-studio  
lingua-libre  
wikimedia-commons  
git  
github

## 3. "Try it out" links

**PASTE** · Three links: the code, the installable demo APK, and our Premium design. Use 'add another link' for each extra one.

<https://github.com/cypher-30/Root>  
<https://github.com/cypher-30/Root/releases/tag/v0.1.0-shipaton>  
<https://github.com/cypher-30/Root/blob/main/docs/PREMIUM.md>

#### 4. Image gallery

**UPLOAD** · Upload in this order. The project thumbnail is a separate field; use Motion\exports\root-thumbnail.jpg there. All are PNG and under 1 MB. 01 to 05 are 3:2.

uploads\01-root-closer-to-home.png  
uploads\02-root-a-session-not-a-streak.png  
uploads\03-root-a-path-never-a-locked-door.png  
uploads\04-root-real-voices-and-culture.png  
uploads\05-root-premium-sets.png  
uploads\06-root-app-icon-1024.png  
uploads\07-root-app-screenshot-1179x2556.png

#### 5. Video demo link

**FILL IN** · Required. Under 2 minutes, public, showing Root running on a real Android device. See Upload-Guide.txt.

[PASTE YOUR PUBLIC YOUTUBE OR VIMEO URL]

ADDITIONAL INFO (for judges and organizers)

#### 6. Did you attach a 1024 x 1024 uncropped image of your app icon?

**SELECT** · Only after uploading 06-root-app-icon-1024.png.

Yes

#### 7. Did you attach a screenshot of your app WITHOUT device frames?

**SELECT** · Only after uploading 07-root-app-screenshot-1179x2556.png.

Yes

#### 8. Was the first version of your app released on the App Store, Google Play Store, or Samsung Galaxy Store between August 1 and September 30, 2026?

**SELECT** · Root is not on any store. Next Gen doesn't require it.

No

#### 9. Are you an employee of RevenueCat or a Shipaton Sponsor?

**FILL IN** · Answer truthfully for the whole team.

[No, if that is true for everyone on the team]

#### 10. What type of app did you build?

**SELECT** · Tick Android only.

Android

#### 11. URL to your published iOS or Mac app on Apple's App Store

**LEAVE BLANK** · Not published.

## 12. URL to your published Android app on the Google Play Store

LEAVE BLANK · Not published.

## 13. URL to your published app on the Samsung Galaxy Store

LEAVE BLANK · Not published.

## 14. (Next Gen Only) URL to your code repository

PASTE · The default branch (main) holds the current build.

`https://github.com/cypher-30/Root`

## 15. (Next Gen Only) Student/academic email address

FILL IN · Must be a real academic address. Eligibility may be checked against JetBrains/swot.

[YOUR STUDENT EMAIL]

## 16. (Next Gen Only) Minor entrant confirmation

FILL IN · Tick if nobody on the team is a minor, or if a parent or guardian has completed the consent form. The form linked here and the one in the rules differ; if a minor is involved, check with the organizers.

[Tick only if it is true]

## 17. RevenueCat project ID

FILL IN · RevenueCat dashboard > Project settings. This is the project ID, not an API key. Never paste an API key here.

[YOUR REVENUECAT PROJECT ID]

## 18. Promo code for unlocking premium features

PASTE · Paste only the code. Judges enter it under Have a code? on the Premium screen (Profile > Root Premium, or See Premium on any locked set in Explore). It opens every premium set in every language on their device.

SHIPATON2026

## 19. For Grand Prize Award: growth since launch

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 20. Build in Public Award: how building in public helped

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 21. Build in Public Award: links

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 22. HAMM Award: monetization model

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 23. RevenueCat Peace Prize

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 24. RevenueCat Design Award

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 25. Catvertising Award

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 26. Best Game Award

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 27. Which Influencer Award category are you entering?

LEAVE BLANK · Leave it on 'Make a selection'.

## 28. Influencer Award description

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 29. Ship Kotlin Everywhere Award

LEAVE BLANK · Root is native Android, not Kotlin or Compose Multiplatform, and isn't published.

## 30. Most Viral App: Noise account email

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 31. Most Viral App: how you used Noise

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 32. Best App for Galaxy Award

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 33. Idea to Income: Replit preview URL and username

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 34. Idea to Income: how Replit helped

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 35. Keep Them Coming Back: OneSignal App ID

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 36. Keep Them Coming Back: how you used OneSignal

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 37. The Growth Loop Award (Layers)

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 38. Funnel Vision: RevenueCat Funnel URL

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 39. Funnel Vision: Stripe Project ID

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 40. Funnel Vision: funnel design and performance

LEAVE BLANK · Not entering: we are entering Next Gen only.

## 41. Do you have any additional notes for the judges?

**PASTE** · Honest and short. Judges see this; the public page doesn't.

Root is a Next Gen (student) submission for Android.

The code is in our public, MIT-licensed repository: <https://github.com/cypher-30/Root>

To try it: install the demo APK from our GitHub release, <https://github.com/cypher-30/Root/releases/tag/v0.1.0-shipaton> (Android 8.0 or later; allow "install unknown apps"; no account). It is a debug-signed demo build with the RevenueCat Test Store connected, so purchases are sandbox and cost nothing. To build it yourself: `.\gradlew.bat :app:installDebug`. No API key is needed to practise, learn, explore, add your own words or record. The RevenueCat paywall needs a Test Store key in user-level `gradle.properties`; the README explains this.

Where things stand today:

- RevenueCat is integrated with per-language entitlements (`premium_<language>`), an all-languages bundle (`premium`), restore, and an offline entitlement cache.
- Premium is two extra phrase sets per language (US\$4.99 one language, US\$9.99 all, one-time). They show locked in Explore; "See Premium" opens the paywall, which lists that language's sets.
- Purchases run through RevenueCat's Test Store (sandbox, no real money). Root is not on Google Play yet.
- To open everything without buying, enter the code SHIPATON2026 on the Premium screen (Profile > Root Premium, or See Premium on any locked set). Remove code locks the sets again. The code is local to the device and is not a RevenueCat transaction.
- All phrases, free and premium, are source-checked against public references but not yet reviewed by native speakers, and the app says so. Six phrases play genuine, credited Lingua Libre recordings.

Thank you for looking at Root. We built it for everyone who grew up far from the place their language lives.

## 42. Are you interested in RevenueCat Growth Fund and having your app considered for it?

**FILL IN** · Your call. There is no right answer here.

[Your choice]

Story length: 1060 words.